Proof:

$$\begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \\ -1 & -4 & -2 \end{bmatrix} \xrightarrow[-2\times \text{Row}1]{\text{Row}2=\text{Row}2} \begin{bmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 3 & 12 & 6 \\ -1 & -4 & -2 \end{bmatrix} \xrightarrow[-3\times \text{Row}1]{\text{Row}3=\text{Row}3} \begin{bmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & -4 & -2 \end{bmatrix}$$

$$\xrightarrow[+\text{Row}1]{\text{Row}4=\text{Row}4} \begin{bmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

. It's easy to see that only column 1 is the pivot column. By Theorem 13, column 1 is the basis of Col A. By Defn, dimension of the column space of the matrix is 1. By Defn. of rank, rank of the matrix is 1. □

25) Let A be an $n \times p$ matrix whose column space is p-dimensional. Explain why the columns of A must be linearly independent.

Soln: By defn, column space of A has a basis consisting of p vectors each comprising of n entries. By example 4 in Sec-2.8, we know Col A is a subspace of $\mathbb{R}^n$. Since there are p columns of A and Col A is the set of all linear combinations of A, by Theorem 15, the columns of A must be linearly independent. □

26) Suppose columns 1, 3, 5 and 6 of a matrix A are linearly independent (but are not necessarily pivot columns) and the rank of A is 4. Explain why the four columns mentioned must be a basis for the column space of A.

Soln: By defn, rank A = dim Col A = 4. By defn, the no. of vectors in the basis of Col A is 4. By example 4 in Sec-2.8, we know Col A is a subspace of the no. of rows of A. By Theorem 15, columns 1, 3, 5 and 6 of A form a basis for the column space of A. □

27) Suppose vectors $b_1, \ldots, b_p$ span a subspace W, and let $\{a_1, \ldots, a_q\}$ be any set in W containing more than p vectors. Fill in the details of the following argument to show that $\{a_1, \ldots, a_q\}$ must be linearly dependent. First, let $B = [b_1 \ldots b_p]$ and $A = [a_1 \ldots a_q]$.

a) Explain why for each vector a_j, there exists a vector c_j in $\mathbb{R}^p$ such that $a_j = Bc_j$.

Soln: We know $a_j \in W$. From defn. of Span, $\exists\, c_1, c_2, \ldots, c_p \in \mathbb{R}$ s.t.

$$a_j = c_1 b_1 + c_2 b_2 + \ldots + c_p b_p$$